

continuous monitoring are essential for safeguarding Industrial IoT environments, but these are often overlooked.

Lack of security as an end goal:

Security should not be an afterthought but an integral part of the entire process. Neglecting security for the sake of convenience or expediency can lead to vulnerabilities.

As organisations forge ahead with digital transformation, they must prioritise security in tandem with innovation. A robust security strategy, modernisation of legacy systems, and ongoing monitoring and improvement efforts are critical for protecting Industrial IoT assets. By addressing these challenges proactively, organisations can navigate the perfect storm of threats in the digital transformation era and emerge stronger.

In this article, we delve into the strategies and principles that can fortify your IIoT security posture.

Pillars and automation in IoT security

To build a robust IIoT security framework, focus is required on three key pillars: people, process, and

technology. Recognising that humans are often the weakest link in security, investment should be done in security awareness training, gamify it for employee engagement, and conduct thorough background checks for new hires. Security by design is vital. It is necessary to establish clear processes, governance, and a comprehensive security strategy.

One needs to define the security architecture, including essential security controls such as firewalls, endpoint security, SIEM (security information and event management), and OT security tools. It is important to cultivate a security-conscious culture, not only within your organisation but also among your partners and vendors.

Despite diligent security efforts, modern IIoT environments still face formidable challenges:

Lack of visibility: OT teams may not possess the technical know-how to identify security issues, leading to oversight.

Zero-day attacks: Rapidly evolving threats, such as zero-day attacks, can exploit vulnerabilities before security measures are in place.

Overwhelmed security operations:

Security teams may struggle to keep pace with the sheer volume of threats and alerts.

Complexity: As IIoT environments expand, managing and securing an increasingly complex ecosystem becomes challenging.

Securing IIoT environments demands a multifaceted approach. The convergence of people, processes, and technology is essential. Embrace a defence-in-depth approach, adhere to industry standards, and consider the zero trust security model. By implementing these principles and remaining vigilant, organisations can navigate the IIoT security landscape. In this article, we explore the importance of continuous security inspection, the role of automation, and strategies to enhance IoT security.

Continuous security inspection involves vigilant monitoring of your IoT ecosystem for signs of intrusion or compromise. Whereas, automation helps streamline security processes, reduce response times, and enhance overall protection. Here are some key components of automation in IoT security:

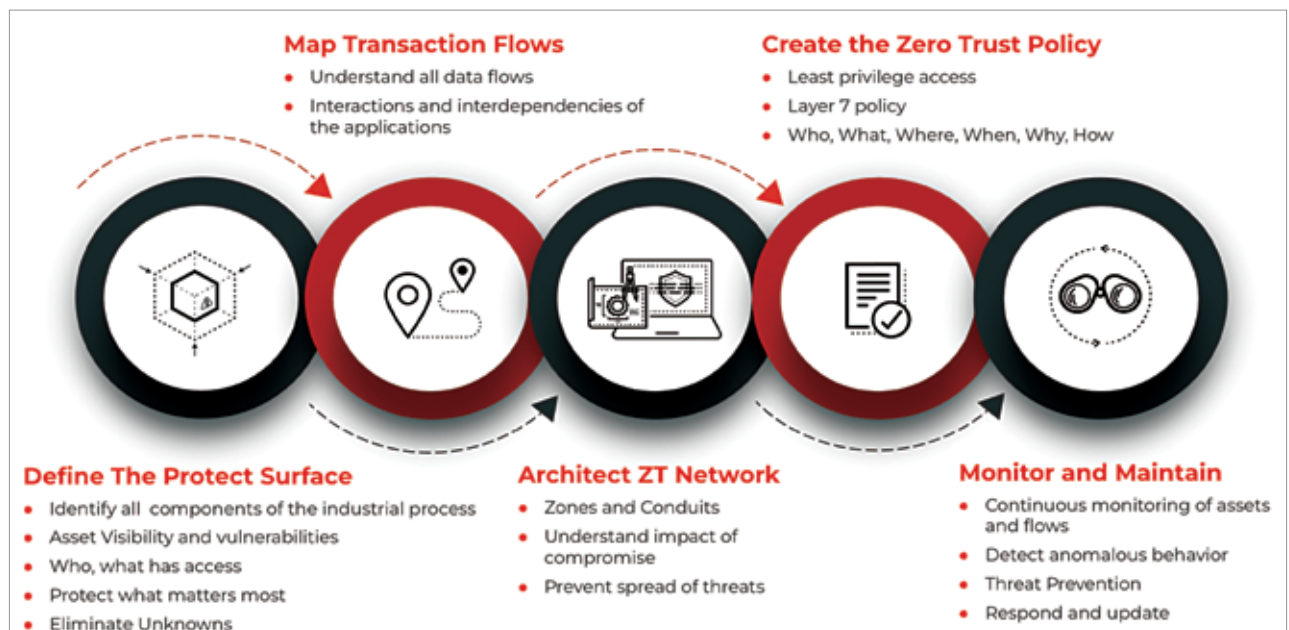


Figure 5: Zero trust security model